

Key Formulae

Population

Mean

$$\bar{y} = \left(\frac{1}{n}\right) \sum_{i=1}^n y_i$$

Variance

$$S^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2$$

SRS

Sampling Variance to the Mean

$$V(\bar{y}) = \frac{1}{n} \sum_{s=1}^S (\bar{y}_s - \bar{Y})^2$$

$$se(\bar{y}) = \sqrt{V(\bar{y})}$$

Computed Sampling Variance to the Mean

Base Formula

$$V_{SRS}(\bar{y}) = \frac{1-f}{n} S^2$$

Binary Variable

$$v_{SRS}(p) = \left(\frac{1-f}{n}\right) p(1-p)$$

Search for Sample Size

$$n = \frac{(1-f)S^2}{(se(\bar{y}))^2}$$

Design Effect

$$d^2 = \frac{V(\bar{y})}{V_{SRS}(\bar{y})}$$

Sampling Fraction

$$f = \frac{n}{N}$$

Stratified Sampling

Strata mean

$$\bar{y}_{st} = \sum_{h=1}^n w_h \bar{y}_h$$

Stratified Sampling variance of the Mean

$$v(\bar{y}_{st}) = \sum_{h=1}^3 w_h^2 v(\bar{y}_h)$$

PROBLEM 3

a. The sample size is computed as

$$n = \frac{(1-f)S^2}{(se(\bar{y}))^2}$$

where $s^2 = 2.7^2 = 7.29$, $se(\bar{y}) = 0.15$, and f is assumed to be close to 0 since we have a large population. Therefore:

$$n = \frac{7.29}{(0.15)^2} = 324$$

b. To achieve $se(\bar{y}) = 0.15$, which is the same standard error of the mean as in a), the sample size that is necessary for the proportionate stratified design will be smaller than that computed in a). This is because, with the same sample size, proportionate stratified design leads to a smaller sampling variance or standard error than simple random design.

Example Case

1. Consider a small population consisting of $N = 8$ students with the following exam grades:

$$Y_1 = 72, Y_2 = 74, Y_3 = 76, Y_4 = 77, Y_5 = 81, Y_6 = 84, Y_7 = 85, Y_8 = 91$$

a. Compute the population mean $\bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i$

Answer:

$$\bar{Y} = \frac{1}{8} (72 + 74 + 76 + 77 + 81 + 84 + 85 + 91) = \frac{555}{8} = 80$$

b. Compute the population variance $S^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2$

Answer:

$$S^2 = \frac{1}{8-1} ((72-80)^2 + (74-80)^2 + (76-80)^2 + (77-80)^2 + (81-80)^2 + (84-80)^2 + (85-80)^2 + (91-80)^2) = \frac{288}{7} = 41.143$$

c. Mean of each possible sample of size 2 ($\bar{y}_s$):

$$\bar{y}_1 = \frac{1}{2} (y_1 + y_2) = 73$$

:

$$\bar{y}_{28} = \frac{1}{2} (y_7 + y_8) = 88$$

d. Sampling variance of the sample means obtained in c):

$$V(\bar{y}) = \frac{1}{28} \sum_{s=1}^{28} (\bar{y}_s - 80)^2 = 16$$

Computed/estimated sampling variance:

$$V_{SRS}(\bar{y}) = \frac{1-f}{n} S^2 = \frac{1-1/4}{2} 41.143 = 15.429$$

4. The frame for an education survey includes $A = 2000$ high school, each containing $B = 1000$ students. An epsem sample of $n = 3000$ students is selected in two stages. At the first stage, $a = 100$ schools are selected randomly, and at the second stage $b = 30$ students are selected in each sampled school. Of the selected students, 30% ($P = 0.3$) reported having access to computers at home. A published estimate gives the standard error of this percentage as $se(\bar{y}) = 1.4\%$. Ignoring finite population corrections and approximation $(n-1)$ by n , estimate the following:

A. The design effect d^2 for the sample percentage.

Answer:

$$se(\bar{y}) = \sqrt{V(\bar{y})} \rightarrow V(\bar{y}) = (se(\bar{y}))^2 = 0.014^2 = 0.000196$$

$$V_{SRS}(\bar{y}) = \frac{1-f}{n-1} P(1-P) \rightarrow \frac{1}{n} P(1-P) = \frac{1}{3000} * 0.3 * 0.7 = 0.00007$$

$$d^2 = \frac{V(\bar{y})}{V_{SRS}(\bar{y})} = \frac{0.000196}{0.00007} = 2.8$$

5. A stratified sample was selected to estimate annual visits to a doctor. The stratum size (N_h), the proportion of the population (W_h), The sample size (n_h), the sampling fraction (f_h), the total number of doctor visits (y_h), the mean number of doctor visits ($\bar{y}_h$), the sample element variance (S_h^2) in each of the three strata are as follows:

Stratum	N_h	W_h	n_h	f_h	y_h	$\bar{y}_h$	S_h^2
Young	3200	0.40	192	0.06	1152	6	5
Middle Aged	4000	0.50	240	0.06	1200	5	4
Old	800	0.10	48	0.06	384	8	7
Total	8000	1	480	0.06	2736		

a. Compute the unweighted mean of the strata means.

$$\bar{y} = \frac{6 + 5 + 8}{3} = 6.333$$

b. Compute the weighted stratified mean.

$$\bar{y}_{st} = 0.4 * 6 + 0.5 * 5 + 0.1 * 8 = 5.7$$

c. Compute the sampling variance of the mean from part b.

$$V(\bar{y}_{st}) = \sum_{h=1}^3 W_h^2 * V(\bar{y}_h) = W_1^2 \left[\frac{1-f_1}{n_1} \right] S_1^2 + W_2^2 \left[\frac{1-f_2}{n_2} \right] S_2^2 + W_3^2 \left[\frac{1-f_3}{n_3} \right] S_3^2$$

$$= 0.4^2 \left[\frac{1-0.06}{192} \right] 5 + 0.5^2 \left[\frac{1-0.06}{240} \right] 4 + 0.1^2 \left[\frac{1-0.06}{48} \right] 7$$

$$= 0.009204$$

d. Compute a 95% confidence interval for the mean computed in part b.

$$se(\bar{y}_{st}) = \sqrt{V(\bar{y}_{st})} = \sqrt{0.009204} = 0.09594$$

$$95\% CI \rightarrow \alpha = 0.05$$

$$\bar{y}_{st} \pm Z_{1-\frac{\alpha}{2}} * se(\bar{y}_{st}) = \bar{y}_{st} \pm Z_{0.975} * se(\bar{y}_{st}) = 5.7 \pm 1.96 * 0.09594 = 5.7 \pm 0.188$$

$$\rightarrow [5.512; 5.888]$$